



Wolkite university

College Of Computing and Informatics

Department of Computer Science

Course Title: Web Programming

Course Code: CoSc 3082

Credit Hrs.: 4 ECTS: 7 Lecture Hrs.: 3

Lab Hrs.: 3 Tutorial Hrs. 1

Targeted Department: Computer Science

Instructor Name: Ayele G.

email: - ayele.gobezie@wku.edu.et

Course Category: Compulsory

Academic year 2023/2024

Year III Semester II

Course Description

Client-server Architecture: Web page Design and development; information Architecture & visualization; static & dynamic pages, client-side programming using scripting languages (JavaScript), OOP programming using PHP, File manipulation and Management using PHP. Multimedia web programming, dynamic web content creation; session tracking; Internet protocols, PHP Web controls, connecting web page to data base, loading and manipulating XML documents, web security; cryptography theory.

Objectives

The objective of this module is to familiarize students:

- ✓ Provide fundamental concepts and skills for web-based applications.
- ✓ Construct Web sites using HTML and Server-Side scripting Languages.
- ✓ Implement client-side interactivity
- ✓ Use CSS to manage Web site presentation and maintenance

- ✓ Select and customize existing technologies to set up and maintain web servers
- ✓ Specify, build and manage form and content of information-rich web sites
- ✓ Design, implement and evaluate client-server systems following specific protocol specifications, taking into account concurrency issue.

Course outline

Chapter 1: The world Wide Web (Week 1)

- 1.1. Course overview
- 1.2. The Internet and World Wide Web
- 1.3. Web Hosting and Domain Name registration

Chapter 2: Hyper Text Markup Language (HTML) (Week 2& 3)

- 2.1. Introduction to HTML
- 2.2. HTML tags
 - 2.2.1. Basic HTML tags (HTML, HEAD, BODY, TITLE)
 - 2.2.2. Meta tag
 - 2.2.3. HTML Comments
 - 2.2.4. HTML Link
 - 2.2.4. HTML Text Formatting tags
 - 2.2.5. HTML image inserting tag
 - 2.2.5.1. IMG tag and its attribute
 - 2.2.5.2. Inserting Image Map
 - 2.2.6. HTML Table
 - 2.2.7. Ordered and Unordered List in HTML
 - 2.2.8. HTML Frames
 - 2.2.8.1. Frame Set
 - 2.2.8.2. Internal Frame
 - 2.2.9. HTML Form and Form Controls
 - 2.2.9. Inserting Multimedia in HTML
 - 2.2.9.1. Embed vs Video and Audio tags
 - 2.2.10. HTML Graphics
 - 2.2.10.1. HTML Canvas
 - 2.2.10.2. HTML SVG

Chapter 3: Cascading Style Sheet (CSS) (Week 4& 5)

- 3.1. CSS Basics
 - 3.1.1. Introduction to CSS
 - 3.1.2. CSS Syntax (CSS Selectors and Declarations)

- 3.1.3. Attaching CSS with HTML (External, Embedded and Inline)

3.2. Style Sheet Rules

- 3.2.1. Style Inheritance
- 3.2.2. Style Rules Precedence

3.3. Style Properties

- 3.3.1. Foreground and Background Properties
- 3.3.2. Font and Text Properties
- 3.3.3. CSS Box Model
- 3.3.4. Table Styling Properties
- 3.3.5. More on Styling List (Creating Navigation bars)
- 3.3.6. Layout and Positioning Properties

3.4. CSS Measuring Units

Mid Exam (Week 5)---20%

Chapter 4: Client-Side Scripting (JavaScript) (Week 6 & 7)

- 4.1. Introduction to JavaScript
- 4.2. JavaScript Basic
 - 4.2.1. JavaScript Syntax
 - 4.2.2. Attaching JavaScript to HTML (External, Embedded, Inline)
- 4.3. JavaScript Comments
- 4.4. Basic JavaScript Input Output
- 4.5. JavaScript Data Types and Variables
 - 4.5.1. JavaScript Data types
 - 4.5.2. Variable declaration in JavaScript
 - 4.5.3. Data Type Conversion
- 4.6. Arithmetic and Logical Operators in JavaScript
- 4.7. Control Structures (Conditional and Looping Statements)
- 4.8. Array in JavaScript
- 4.9. JavaScript Functions
- 4.10. JavaScript DOM (Document object Model)
 - 4.10.1. Accessing HTML elements in JavaScript
 - 4.10.2. CSS in JavaScript

- 4.10.3. Events in JavaScript
- 4.10.4. Handling Exception in JavaScript
- 4.11. Form Processing using JavaScript
- 4.12. JavaScript BOM (Browser Object Model)
 - 4.12.1. JavaScript Window
 - 4.12.2. JavaScript Location
 - 4.12.3. JavaScript Location
 - 4.12.4. JavaScript Cookies

Week 7 –Quiz (5%)

Chapter 5: Server-Side Scripting (PHP)(Week 8& 9)

- 5.1. Introduction to PHP
- 5.2. Basic PHP Syntax
 - 5.2.1. PHP Comments
 - 5.2.2. Predefined and User Variables in PHP
- 5.3. PHP Output Statements
- 5.4. Data Types and Variables in PHP
- 5.5. Arithmetic and Logical Operators
- 5.6. Conditional Statements
- 5.7. Loop Statements in PHP
- 5.8. Arrays in PHP

- 5.9. PHP Functions
- 5.10. Form Processing using PHP
- 5.11. PHP File Upload
- 5.12. PHP Cookies and Session
- 5.13. Database Programming using PHP
 - 5.13.1. Overview on MySQL database
 - 5.13.2. Creating Database Connection in PHP
 - 5.13.3. Sending Query to MySQL Database using PHP
 - 5.13.4. Processing Query Result.
- 5.14. PHP File Input-Output
- 5.15. PHP Date and Time
- 5.16. PHP Mathematical Functions
- 5.17. PHP OOP

Chapter 6: Advanced JavaScript and XML (AJAX) (Week 11)

- 6.1. Introduction to AJAX
- 6.2. XMLHttpRequest Object
- 6.3. Sending Request to PHP server
- 6.4. Handling Response from Server

Teaching - Learning methods

- ✓ The Course will be offered through lectures, lab and Students do home assignments & projects.

Assessment method: - Quiz (5%) + Mid (20%) + Project(15%)+Assignment/Presentation(10%)=50% and final exam 50%

Reference books:

- ✓ Fundamental Concepts for Web Development: HTML5, CSS3, JavaScript and much more, by Roxane Anquetil
- ✓ Web Design with HTML, CSS, JavaScript and jQuery Set 1st Edition by Jon Duckett (Author)
- ✓ W3Schools, <https://www.w3schools.com>

NB:

- ✓ Every student should attend all the sessions (Lecture and Lab)
- ✓ Students should group themselves in to 3 for project and assignment work and identify their project titles on the first day of the class
- ✓ Students should submit every assignment and project according to the deadline.
- ✓ Students should demonstrate their work (Project, Assignments)
- ✓ Students should sit for the examinations